EEG

6.2 **Exploratory Statistics**: Boxplot comparing tasks Rest vs Task for each frequency channel, a heatmap of each power variable by channel, and a bar plot of the aver. age power by brain region. While doing this task, answer the following questions:

Which frequency band increases during the task?

Mirando el boxplot que armé al principio de 6.2: las bandas que muestran la caja de Task claramente desplazada hacia arriba respecto a la de Rest son **BetaPower** y **GammaPower**. **DeltaPower** también muestra un leve desplazamiento hacia arriba, aunque menos pronunciado

Which brain region shows the strongest task-related change?

To answer this, I first calculated the change between Task and Rest for the three bands tied to my hypothesis: Alpha, Beta, and Gamma. I left out Delta and Theta on purpose, since they are not part of what I am trying to test, and including them would just add noise to the result.

I looked at this in two ways: absolute change and percent change. Absolute change keeps everything in the original units of power, while percent change expresses the change relative to each channel's own Rest baseline. These two methods do not always agree, and the difference matters here. Because Gamma has a much smaller baseline power than the other bands, absolute change tends to minimize how much Gamma actually moves. In relative terms, however, Gamma is the band that changes the most. In fact, the single largest jump I found anywhere in the dataset is GammaPower in the Centroparietal region (+74.5%), and that result is only visible once I look at percent change. Without it, this jump would have gone unnoticed, since in absolute terms it looks unremarkable. This makes sense biologically too: Gamma power is the band that reacts the most when a person switches from rest to actively performing a task.

To get one combined score per region, I summed the absolute value of the change across Alpha, Beta, and Gamma. Summing the absolute value matters here, since a strong drop in Alpha and a strong rise in Beta could otherwise cancel each other out and hide how much that region actually changed.

In Centroparietal, the percent change was -24.4% for Alpha, +30.1% for Beta, and +74.5% for Gamma, giving a combined score of 129.0%, the highest of any region (the second-highest, Temporal, reached 92.9%). Using absolute change instead, Centroparietal still comes out on top with a combined score of 7.05 (compared to 5.82 for the second-highest region, Occipital).

Using both methods, the **Centroparietal region** shows the strongest task-related change overall.

After that, I realized that the way to solve this problem of the different waves (Delta, Theta, Alpha, Beta, Gamma) having naturally different magnitudes, where comparing their changes directly in original units biases the comparison in favor of the waves with larger numbers, is by using the z-scores I made in 6.1. This puts all the waves on a common scale (standard deviations relative to their own typical behavior).
What the standardized power change tells us is "how much the band changed, relative to how consistent or dispersed its normal behavior is." A band with a lot of natural variability between channels (as could happen with Gamma, which is noisier) needs a bigger change for that change to register as "significant" in z terms, compared to a more stable band.
Using this third method, Centroparietal is still the region that showed the strongest change compared between rest and task.

Does alpha power decrease during cognitive activation? yes **Does beta or gamma power increase?** yes

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6.7 **Interpretation**: write a short interpretation answering:

Which EEG features best distinguish Rest from Task? Is alpha suppression observed?

Is beta/gamma activation observed?

Entropy, BetaPower, and AlphaPower are the features that best distinguish Rest from Task, according to the model. The positive sign of Entropy and Beta means that higher values push the prediction toward "Task". The negative sign of Alpha means that higher values push the prediction toward "Rest."

Although Gamma showed the largest relative change (percent change) during Task in the exploratory analysis (6.2), its coefficient in the logistic regression model is modest, likely because Entropy captures overlapping information (correlation = 0.52 between Gamma_z and Entropy_z). When viewed in isolation (percent change), Gamma is highly reactive; but within the full model, much of its discriminative power is shared with Entropy.

Yes, alpha suppression is observed. Both the descriptive analysis (the decrease in the median of AlphaPower between Rest and Task in the boxplot) and the negative coefficient of AlphaPower_z in the classification model are consistent with this classic phenomenon reported in the EEG literature.

Both Beta and Gamma show clear activation during Task. In terms of relative change (percent change), Gamma actually shows the largest proportional increase of any band, particularly in the Centroparietal region (+74.5%). However, Gamma's coefficient in the logistic regression model is modest, not because Gamma doesn't change, but because it is highly correlated with Entropy (r = 0.52), which captures overlapping information. Within the model, much of Gamma's discriminative power is shared with Entropy, causing the model to assign most of the weight to Entropy rather than Gamma.

Which brain regions contribute most strongly?

If "contributing most strongly" means which regions provide the most additional predictive power, then frontocentral shows the largest positive coefficient (0.45), followed by Parietal (0.23). However, region contributes secondary information to the model. All region coefficients are considerably smaller than the band coefficients (Entropy: 1.79, Beta: 1.18, Alpha: -1.00), suggesting that the spectral pattern carries most of the discriminative power, and the region refines the prediction only at the margin.

Does the model support the hypothesis that cognitive activation changes EEG spectral structure?

Yes. A logistic regression classifier trained on standardized EEG frequency-band features and brain region got 93.8% accuracy, 100% sensitivity, and an AUC of 0.984, indicating that the spectral pattern shifts detectably between Rest and Task. This is also supported by the specific features the model weighted most heavily: Entropy and BetaPower carry large positive coefficients, meaning higher cognitive complexity and faster-frequency activity push the prediction toward

Task, while AlphaPower carries a large negative coefficient, meaning higher slow-wave resting activity pushes the prediction toward Rest.